

Chanyeok Choi

Robot Engineer · MS in Applied AI, Hanyang University

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Profile

Robot engineer working on **critical-phase detection** for vision-language-action policies (preprint, under review). Broader interests include safe and generalizable robot learning, multi-agent reinforcement learning, and sim-to-real transfer. Previously built field-deployed service robots (auto-climbing mopping platform, autonomous delivery platform) — most of my intuition about real-world control and edge cases comes from there.

Education

Hanyang University , Applied Artificial Intelligence - Advisor: Prof. Youngmoon Lee - Research on safety-RL, poisoning attacks on multi-agent RL, and vision-language-action policy failure detection.	Seoul, South Korea Feb 2022 – present
Hanyang University , Robotics Advisors: Prof. Youngmoon Lee, Prof. Taejoon Park, Prof. Changsoo Han	Seoul, South Korea Mar 2018 – Feb 2022

Experience

Hanyang University , Graduate Research Assistant Robot learning research under Prof. Youngmoon Lee.	Seoul, South Korea Mar 2022 – present 4 years 3 months
Critical-Phase Detection (CPD): zero-shot failure detection for frozen π_0 / $\pi_{0.5}$ VLA policies on LIBERO manipulation. Leave-one-out F1 0.86–0.89 with no per-task reward (preprint, under review). Poisoning attacks on multi-agent RL: adversarial reward-poisoning study on multi-agent control, including taxi ride-sharing systems. Safety-RL: bridging the sim-to-real gap by allowing safe optimization and exploration during robot learning. Vision (HumanPose): hybrid PoseNet/SegNet approach for crowded scenes with overlap and occlusion.	
KITECH — Process Platform Research Division , Research Intern	South Korea Mar 2021 – Aug 2021 6 months
Researched AI methods for detecting defective products on high-speed conveyor belts.	
Hanyang University — CAI Lab , Research Intern	Seoul, South Korea July 2019 – Feb 2020 8 months
IKEA Furniture Assembly Collaborative Robot: RL simulation in which two manipulators identify randomly placed furniture parts and assemble them collaboratively.	

Publications

Critical-Phase Detection for Vision-Language-Action Policies

Zero-shot per-step failure detector for frozen π_0 / $\pi_{0.5}$ VLA policies.

Chanyeok Choi, Youngmoon Lee

angledsugar.github.io/projects/0_cpd

Poisoning Attacks on Multi-Agent Reinforcement Learning Systems

Chanyeok Choi, Jaehwan Cho, Youngmoon Lee

HAPTics: Human Action Prediction in Real-time via Pose Kinematics

Niaz Ahmad, Saif Ullah, Jawad Khan, Chanyeok Choi, Youngmoon Lee

Causes and Fixes of Unexpected Drone Shutoffs

Hojun Choi, Chanyeok Choi, Youngmoon Lee

Snapbot: Enabling Dynamic Human Robot Interactions for Real-Time Computational Photography

Chanyeok Choi, Youngmoon Lee

Leveraging Keypoints as Dynamic Centroids for Unified Representation of Human Pose and Instance Segmentation

Niaz Ahmad, Jawad Khan, Chanyeok Choi, Youngmoon Lee, Kang G. Shin

SSK-DNN: Semantic and Sentiment Knowledge for Incremental Text Sentiment Classification

Jawad Khan, Niaz Ahmad, Chanyeok Choi, Saif Ullah, Gyu Rin Kim, Youngmoon Lee

Awards

Information & Communication Planning and Evaluation President Award (2nd place) — SW Talent Festival

Jan 2023

AnimeGAN Filter Photographer & 58 a manipulator that frames and captures stylized portraits under safety-RL constraints.

Capstone Design Award (3rd place) — Hanyang University

Jan 2021

Guide Robot Dog & 58 a full-stack quadruped robot motivated by guide dogs, intended to assist visually impaired pedestrians.

Teaching

Lead Teaching Assistant — Robot Vision System: Hanyang University, Fall 2022 & Fall 2023. Lead other graduate TAs for a class of 30–40 students; developed an autonomous robot training and vision environment in Unity. [\[Materials\]](#)

Teaching Assistant — Robot Learning: Hanyang University, Spring 2022 & Spring 2023. Class of 40–50 students. Designed a Pac-Man RL project and an ultrasonic-sensor self-driving car kit for real-world maze escape. [\[Materials\]](#)

Skills

Robot Learning

Perception

Engineering

Languages

Korean

Native speaker

English

Professional working proficiency

Interests

Research Interests